01 / Gas diffusion & gas transport

Lecture: BBMS2006-2-Gas diffusion and gas transport-23Sept2026.pdf | 19 source pages

Illustrated study notes

Concise explanations / Original figures / Exam reasoning

The whole lecture in two pathways

Environmental pressure → partial pressure → alveolar–capillary diffusion → oxygen carriage in blood → oxygen unloading in tissues.

Tissue CO_2 production → blood transport → pulmonary unloading → expiration.

What should you be able to do?

Topic	Your target	Notes pages
Partial pressure	Apply Dalton and Henry to altitude and diving.	3–5
Diffusion	Use Fick's law to predict changes and disease effects.	6–7
O_2 transport	Separate pressure, saturation, capacity, content and delivery.	8–10
O_2 dissociation curve	Read the sigmoid curve, P_{50} and left/right shifts.	11–13
CO_2 transport	Explain bicarbonate, chloride shift, Bohr and Haldane.	14–16
Application	Practise integrated cases and closed-book recall.	17–18

How to use these notes

- First pass: inspect each original figure → follow the arrows → explain the mechanism aloud.
- Second pass: cover the reasoning box; decide True/False and state the necessary conditions.
- Final pass: attempt the cases on p. 17, then reconstruct the quick sheet on p. 18.

Source and question labels

- Tutorial Q#: original wording from the first PDF. Answers here are reasoned suggestions, not an official answer key.
- Predicted practice P#: self-written questions based on the tutorial's style; not a forecast of actual exam content.
- [Extension]: supporting physiology not fully developed in this lecture; included only where useful for an original question.
- Lecture references use actual PDF page numbers. The tutorial is one page; original question numbering is preserved.

Teaching figures, curves and relevant visible annotations are retained. Decorative logos and section-divider slides are not repeated. The lecture cover says BPHM 2141 and the tutorial header says BMED2006; these notes follow the files you supplied together.

02 / Which tutorial questions belong here?

Source: BBMS 2006 Tutorial.pdf, PDF p. 1 | All 14 questions screened

Question / focus	Relevance	Where to study it
Q1 Compliance ↑: breathing always easier	Other lecture	Lung mechanics, recoil and work of breathing.
Q2 Apex versus base: blood PO_2	Other lecture	Regional V/Q ratios and gravitational effects.
Q3 Normal saturation → normal delivery	Direct	p. 9: Hb, oxygen content and blood flow.
Q4 Altitude, hyperventilation and acidosis	Cross-topic	p. 4: pressure; extension on alveolar gas and acid–base physiology.
Q5 Right shift reduces lung O_2 uptake	Direct	p. 13: affinity, loading and the plateau region.
Q6 Hypoxemia versus hypoxia	Application	p. 10: content, perfusion, oxygen use and compensation.
Q7 Hyperventilation at unchanged FiO_2	Cross-topic	p. 4: extension on ventilation and PAO_2.
Q8 Smallest bronchioles: highest resistance	Other lecture	Parallel airways, total cross-sectional area and resistance.
Q9 Surfactant increases inspiratory work	Other lecture	Surface tension, Laplace's law and lung mechanics.
Q10 CO: saturation and affinity	Direct	p. 13: fewer available sites, greater affinity at remaining sites.
Q11 Chronic hypercapnia: ventilatory drive	Other lecture	Chemoreceptors and respiratory control; CO_2 transport alone is insufficient.
Q12 CA inhibitor and HCO_3^- excretion	Cross-topic	p. 16: CA is in this lecture; renal/drug mechanisms are extensions.
Q13 Diving with unchanged N_2 fraction	Direct	p. 5: Dalton + Henry; total pressure must be considered.
Q14 Equal minute ventilation	Other lecture	Dead space, tidal volume and alveolar ventilation.

What does the question style test?

- Concept substitution: saturation ≠ content ≠ delivery; fraction ≠ partial pressure.
- A mixed statement: the first clause can be correct while the explanation is wrong, as in Q4 and Q10.
- Absolute wording: always, must and unchanged often hide a missing condition or variable.

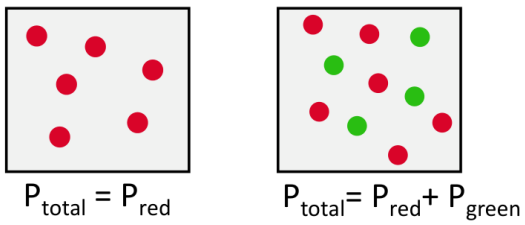
A reusable answer structure

Judgement → identify the key variable → give a short causal chain
→ state the condition or provide a counterexample.

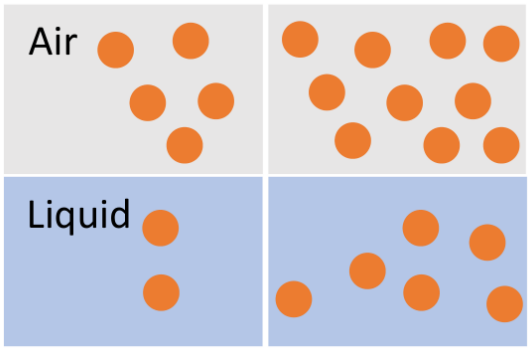
03 / Partial pressure and dissolved gas

Lecture PDF pp. 4–5 | Foundation for Tutorial Q13

Law	Relationship	What does it tell you?
Dalton's law	Pgas = Fgas × Ptotal Ptotal = sum of partial pressures	The pressure contributed by one gas in a mixture.
Henry's law	Cdissolved = α × Pgas	Equilibrium dissolved concentration at a fixed temperature, gas and solvent.



Original: partial pressures add up (lecture p. 4).



Original: gas–liquid equilibrium (lecture p. 5).

Connect the laws

At unchanged gas fraction: Ptotal ↑ → Pgas ↑ → Cdissolved ↑.
An unchanged percentage does not imply an unchanged dissolved concentration.

Situation	Calculation / mechanism	Interpretation
Dry air at sea level	760 × 0.21 = 159.6 mmHg	PO_2 of dry inspired air, not arterial PO_2.
Lecture's 8,000 m example	250 × 0.21 = 52.5 mmHg	Use the pressure supplied by the slide; FiO_2 remains about 21%.
Pressure falls	Pgas ↓ → equilibrium dissolved amount ↓	Gas may leave solution; rapid decompression can form bubbles.

Precision note: solubility versus dissolved concentration

The slide says "greater pressure, greater solubility". More precisely, at fixed temperature and medium, α is unchanged; the equilibrium concentration of dissolved gas rises with partial pressure.

Predicted practice P1 · Self-written; not an original exam question

"If oxygen remains 21% of inspired air at altitude, inspired PO_2 is unchanged."
Reasoning: False. Pgas = Fgas × Ptotal. Atmospheric pressure falls, so PO_2 falls even when FiO_2 is unchanged.

04 / Altitude and hyperventilation

Lecture PDF pp. 4–5 | Tutorial Q4 and Q7 | Equations and acid–base mechanisms below are extensions

The primary altitude effect

Barometric pressure PB ↓ → inspired PIO_2 ↓ → alveolar PAO_2 ↓
→ a smaller O_2 pressure gradient into pulmonary blood → impaired oxygenation.

[Extension] Why can hyperventilation help?

Relationship / condition	Use it to explain...
$PIO_2 = FiO_2 \times (PB - PH_2O)$	Humidified inspired gas; $PH_2O \approx 47$ mmHg at 37°C. Distinguish this from dry-air calculations.
$PAO_2 \approx PIO_2 - PaCO_2/R$	Simplified alveolar gas equation; $R \approx 0.8$ under usual room-air conditions.
Alveolar ventilation ↑ at fixed CO_2 production	$PaCO_2$ ↓ → PAO_2 ↑; FiO_2 need not change.
$CO_2 + H_2O \rightleftharpoons H^+ + HCO_3^-$	An acute fall in $PaCO_2$ lowers H^+ and raises pH: respiratory alkalosis.

Tutorial Q4 · Original tutorial question

Hyperventilation at high altitude helps improve oxygenation even though it causes respiratory acidosis.

Suggested answer: False

- Improved oxygenation: this part can be correct.
- The error is “respiratory acidosis”: acute hyperventilation causes respiratory alkalosis.
- Later renal bicarbonate loss can compensate; it does not rename the primary respiratory process.

Tutorial Q7 · Original tutorial question

Hyperventilation can increase arterial PO_2 even when oxygen concentration in inspired air remains unchanged.

Suggested answer: True under usual physiological conditions

FiO_2 unchanged → alveolar ventilation ↑ → $PaCO_2$ ↓ → PAO_2 ↑ → PaO_2 can rise.
PAO_2 is alveolar; PaO_2 is arterial. A severe shunt or other gas-exchange problem may limit improvement.

Predicted practice P2 · Self-written; not an original exam question

“Hyperventilation raises arterial PO_2 by increasing the fraction of inspired oxygen.”
Reasoning: False. It can raise PAO_2 by reducing $PaCO_2$ while FiO_2 stays unchanged.

Scope: alveolar gas and acid–base physiology are included to explain these questions, not to replace the separate respiratory-control lecture.

05 / Diving: fraction is not partial pressure

Lecture PDF pp. 4–5 and 19 | Tutorial Q13

Tutorial Q13 · Original tutorial question

A diver breathing 78% N2 and 21% O2 will have blood dissolved nitrogen level remains unchanged because the fraction of nitrogen is unchanged.

Suggested answer: False
Depth ↑ → absolute ambient pressure ↑ → $PN_2 = FN_2 \times Ptotal$ ↑
→ equilibrium dissolved N_2 concentration ↑ (Henry's law).
The statement fixes the gas fraction but overlooks total pressure.

[Extension] A simple numerical example

Quantity	Surface: about 1 ATA	About 10 m seawater: 2 ATA
FN_2	0.78	0.78: unchanged
PN_2, simplified dry gas	0.78 × 1 = 0.78 ATA	0.78 × 2 = 1.56 ATA
Equilibrium dissolved N_2	Let this be C	About 2C at unchanged temperature and α

- Use absolute pressure: 2 ATA is approximately twice the surface pressure.
- Uptake takes time. Do not claim that every tissue doubles its dissolved N_2 immediately on descent.
- During ascent, lower pressure reduces the equilibrium amount that can remain dissolved.

Reverse the chain to understand decompression
Slow ascent: permits excess gas to reach the lungs and be eliminated.
Rapid ascent: may allow bubbles to form → decompression sickness.
The key here is the pressure–dissolution mechanism, not the full clinical management.

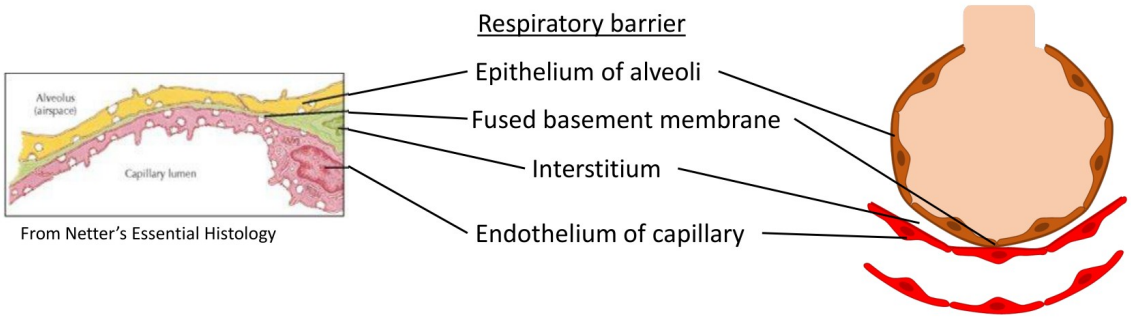
Predicted practice P3 · Self-written; not an original exam question
"At the same depth, reducing FN_2 reduces the equilibrium amount of dissolved nitrogen."
Reasoning: True. At fixed total pressure, FN_2 ↓ → PN_2 ↓ → Cdissolved ↓. State the equilibrium condition.

Before answering any percentage-based gas question

- Check total pressure, humidification and whether gas–liquid equilibrium has been reached.
- Identify the quantity asked: fraction, partial pressure or dissolved concentration?

06 / Diffusion across the respiratory membrane

Lecture PDF pp. 7–8



What determines the rate of diffusion of the gases across the membrane?

Original respiratory barrier, lecture p. 7; source shown on slide: Netter's Essential Histology.

Route and equation

Alveolar gas → alveolar epithelium → thin basement membrane/interstitial region → capillary endothelium → blood.

$\dot{V}gas = A \times D \times \Delta P / T$; $D \propto$ solubility coefficient / $\sqrt{}$molecular weight.

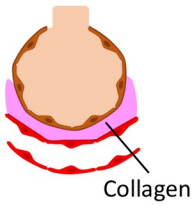
Variable	Direction of effect	Typical application
A: effective surface area	A ↑ → $\dot{V}$ ↑	Emphysema destroys alveolar walls → A ↓.
T: thickness / distance	T ↑ → $\dot{V}$ ↓	Fibrosis thickens the barrier; edema can increase diffusion distance.
ΔP: partial-pressure gradient	ΔP ↑ → $\dot{V}$ ↑	Compare the same gas on opposite sides of the barrier.
D: diffusion coefficient	D ↑ → $\dot{V}$ ↑	Depends on gas and medium; CO_2 diffuses about 20 times more readily than O_2.

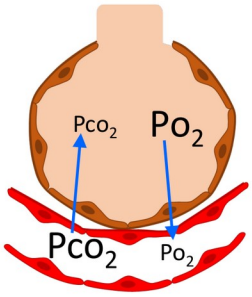
$$V = \frac{A \times D \times \Delta P}{T}$$

V = rate of gas diffusion
A = surface area for diffusion
D = Diffusivity
ΔP = pressure gradient
T = thickness of membrane

CO_2 20x O_2

$$D \propto \frac{solubility}{\sqrt{MW}}$$





Original equation, pressure gradients and thickness illustration, lecture p. 8.

$\dot{V}$ here is gas diffusion rate, not minute ventilation. CO_2 has greater diffusivity mainly because of its greater solubility, despite its larger molecular weight.

07 / Fick's law: blanks, ratios and disease

Lecture PDF pp. 9 and 19 | Original fill-in exercise and visible annotations retained

$$V = \frac{A \times D \times \Delta P}{T}$$

V = rate of gas diffusion
A = surface area for diffusion
D = Diffusivity
ΔP = pressure gradient
T = thickness of membrane

Choose the words to fill in the blank:

Surface area Oxygen Thickness of membrane

Diffusivity Pressure gradient Carbon dioxide

Normally, thickness & diffusivity don't change significantly, therefore the rate of diffusion (V) depends on _____ and _____. In general, _____ is more soluble so it can be eliminated faster, while partial pressure of _____ is high so it can be taken up quickly.

Pathologically, _____ is reduced in emphysema, _____ is increased in fibrosis.

Original "Check your understanding" exercise, lecture p. 9.

Fill-in answers in the original order

Blank / setting	Answer	Reason
Normally relatively stable	Thickness; diffusivity	For the same gas and a healthy barrier, T and D change little.
Main changing determinants	Surface area; pressure gradient	Changes in A and ΔP alter transfer rate.
More soluble / rapidly eliminated	Carbon dioxide	High solubility supports efficient transfer despite a smaller usual ΔP.
Rapid uptake into blood	Oxygen	A favourable alveolar–blood O_2 pressure gradient supports uptake.
Emphysema / fibrosis	Surface area ↓ / thickness ↑	Both impair diffusion capacity.

Predicted practice P4 · Self-written; not an original exam question

"If membrane thickness doubles, diffusion doubles."

Reasoning: False. Other variables unchanged, $\dot{V} \propto 1/T$, so diffusion falls to one half.

Predicted practice P5 · Self-written; not an original exam question

"CO_2 diffuses poorly because its molecular weight is larger than that of O_2."

Reasoning: False. D depends on solubility as well as √MW. The solubility advantage of CO_2 dominates.

A ratio template

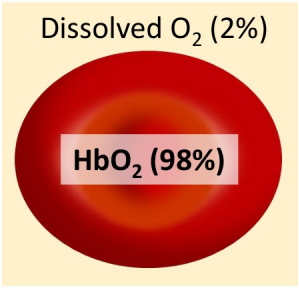
$\dot{V}$new/$\dot{V}$old = (Anew/Aold) × (Dnew/Dold) × (ΔPnew/ΔPold) × (Told/Tnew).

Example: halve A and double T; all else fixed → (1/2) × (1/2) = 1/4 of the original rate.

08 / Five oxygen variables: keep them separate

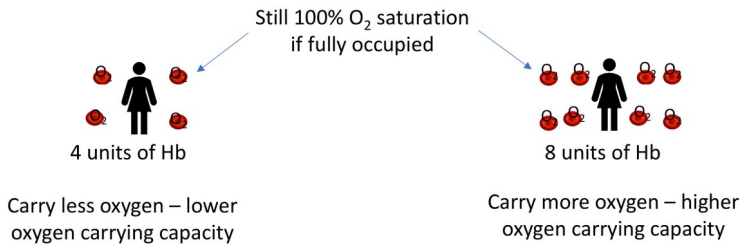
Lecture PDF pp. 11–12 | Foundations for Tutorial Q3, Q6 and Q10

Variable	What it measures	Do not confuse it with...
PaO_2	Partial pressure associated with dissolved O_2 in arterial blood.	Total oxygen content.
SaO_2 / SpO_2	SaO_2: fraction of arterial Hb O_2 sites occupied. SpO_2: a pulse-oximeter estimate.	The amount of Hb present.
O_2 carrying capacity	Maximum Hb-bound oxygen when fully saturated; mainly depends on Hb concentration.	The actual current content.
CaO_2: oxygen content	Total O_2 per volume of arterial blood: Hb-bound oxygen + dissolved oxygen.	PO_2 or saturation alone.
DO_2: oxygen delivery	O_2 delivered per unit time; also depends on cardiac output.	A quantity guaranteed by normal saturation.



Oxygen content

Original: approximately 98% Hb-bound and 2% dissolved O_2, lecture p. 11.



Original: different amounts of Hb can still have the same 100% saturation, lecture p. 12.

- Each Hb molecule has four Fe^{2+}-containing heme groups; each heme can bind one O_2 molecule.
- Dissolved O_2 determines PO_2; the dissociation curve links PO_2 to Hb saturation.

One-line memory aid

Pressure = dissolved-gas pressure; saturation = fraction of sites occupied; Hb = number of sites.
Content = O_2 per dL; delivery = O_2 transported per minute.

The lecture uses SpO_2 broadly for saturation; these notes distinguish SaO_2 from measured SpO_2. The 98%/2% proportions are approximate and condition-dependent.

09 / Normal saturation does not guarantee delivery

Lecture PDF pp. 11–13 | Tutorial Q3 | Quantitative relationships are extensions

Tutorial Q3 · Original tutorial question

A person with normal oxygen saturation must have normal oxygen delivery to tissues.

Suggested answer: False

Normal SaO_2 describes the fraction of available sites occupied, not the total Hb concentration.
Hb ↓ → CaO_2 ↓; cardiac output ↓ → DO_2 ↓ even if CaO_2 is normal.
Counterexamples: severe anemia or low cardiac output with normal saturation.

[Extension] Separate content from blood flow

Equations and units

$CaO_2 \approx 1.34 \times Hb \times SaO_2 + 0.003 \times PaO_2$ (mL O_2/dL).
Hb in g/dL; SaO_2 as a fraction; PaO_2 in mmHg.
DO_2 = cardiac output × CaO_2 × 10 (mL O_2/min, when output is in L/min).

Illustrative case	Hb / SaO_2 / PaO_2	CaO_2	Output / DO_2
Reference	15 / 0.98 / 100	20.0 mL/dL	5 L/min → 1,000 mL/min
Anemia; oxygenation normal	7.5 / 0.98 / 100	10.1 mL/dL	5 L/min → 507 mL/min
Low cardiac output	15 / 0.98 / 100	20.0 mL/dL	2.5 L/min → 500 mL/min

Rounded examples isolate Hb or flow while holding SaO_2 and PaO_2 constant. Hb-bound oxygen capacity ≈ 1.34 × Hb; capacity is not the same as actual CaO_2.

Causal chains worth writing in an answer

- Anemia: Hb ↓ → maximum carrying capacity ↓ → CaO_2 ↓ → DO_2 ↓ if cardiac output is unchanged.
- Low flow: cardiac output ↓ → DO_2 ↓ even with normal oxygen content.
- Normal PaO_2 does not exclude anemia or CO poisoning; it mainly describes dissolved oxygen.

Predicted practice P6 · Self-written; not an original exam question

"Doubling Hb concentration doubles oxygen saturation."
Reasoning: False. Hb concentration affects capacity. Saturation is a proportion and can remain similar at the same PO_2 and curve conditions.

10 / Anemia, cyanosis and tissue hypoxia

Lecture PDF p. 13 | Tutorial Q6 | Hypoxia classification and compensation are extensions

Concept	Key distinction
Anemia	Low Hb concentration → reduced carrying capacity. Causes include iron/B_{12} deficiency and hemolysis.
Cyanosis	Lecture threshold: deoxyHb approximately >5 g/dL. An absolute amount, not a saturation percentage.
Hypoxemia	Low arterial PaO_2.
Tissue hypoxia	Inadequate tissue O_2 supply or use; also depends on perfusion and metabolic demand.



Original skin-colour illustration, lecture p. 13.

The two in-slide True/False questions

- "A person who is anemic must be cyanotic." False: little Hb may mean too little absolute deoxyHb for visible cyanosis, despite poor oxygen delivery.
- "A person who is cyanotic must be anemic." False: cyanosis can occur with normal Hb if oxygenation is poor or local flow is slow.

Tutorial Q6 · Original tutorial question

A person with hypoxemia will have hypoxia; But a person with hypoxia may not have hypoxemia.

Split the statement into its two directions

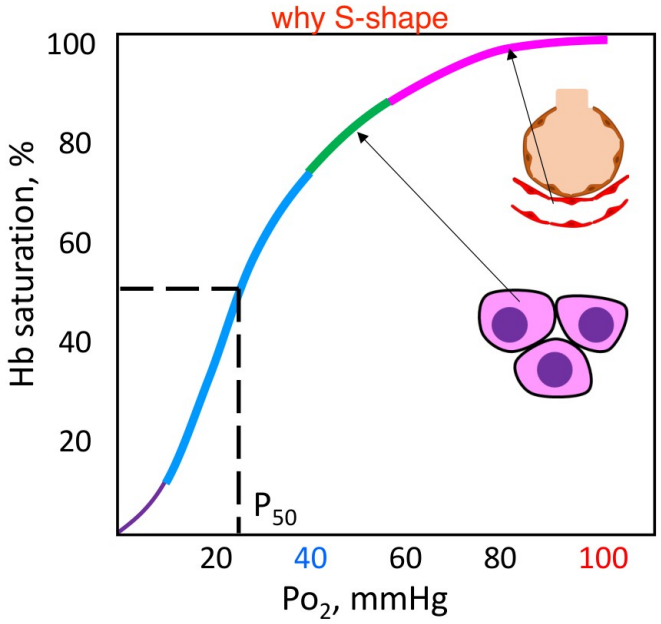
- Second clause: correct. Anemia, low perfusion or impaired cellular O_2 use can cause hypoxia with normal PaO_2.
- First clause: a common introductory simplification, but "will" is too absolute. Mild hypoxemia may be compensated by greater flow or extraction.
- Strict literal reading: the whole statement can be judged False. A simplified course answer may be True; state the compensation caveat. No official key was provided.

Predicted practice P7 · Self-written; not an original exam question

"Absence of cyanosis excludes tissue hypoxia."
Reasoning: False. Severe anemia can compromise delivery without obvious cyanosis.

11 / Read the oxygen dissociation curve

Lecture PDF pp. 14–15



Original sigmoid curve, P_{50} and visible annotations, lecture p. 14.

Why is the curve sigmoid?

Positive cooperativity

First O_2 binds → Hb conformation changes → subsequent O_2 binds more readily.
Unloading is also cooperative: release of one O_2 facilitates further release.

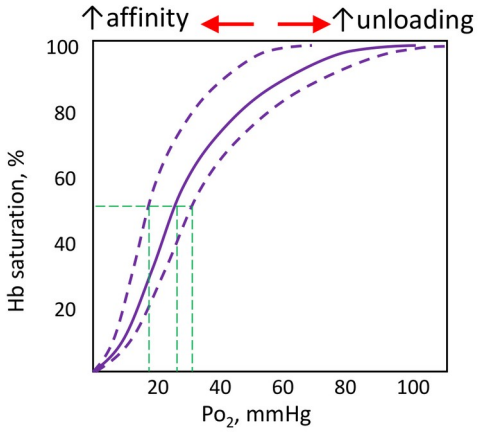
Region / point	Approximate reading	Functional meaning
Pulmonary plateau	PO_2 about 60–100 mmHg.	Moderate PO_2 reductions produce relatively small saturation changes; loading is buffered, not completely invariant.
Steeper tissue region	PO_2 about 15–40 mmHg.	A small PO_2 fall releases a relatively large amount of O_2.
Resting mixed venous blood	$PvO_2 \approx 40$ mmHg; $SvO_2 \approx 75\%$.	About 75% of Hb sites remain oxygenated; PO_2 and saturation are not numerically equivalent.
P_{50}	PO_2 giving 50% Hb saturation; normally ≈ 26.5 mmHg.	Read it from the horizontal axis; it is not “50% of the oxygen pressure”.

Predicted practice P8 · Self-written; not an original exam question
“At PO_2 = 40 mmHg, Hb is normally 40% saturated.”
Reasoning: False. Saturation is approximately 75%; the relationship is nonlinear.

These are approximate standard adult reference values. pH, PCO_2, temperature and 2,3-BPG can alter the curve.

12 / P_{50} and shifts: think about affinity

Lecture PDF pp. 15–16 | Original highlights and handwritten annotations retained



Original P_{50} and curve-shift diagram, lecture p. 15.

Shift	P_{50}	Hb–O_2 affinity	At the same tissue PO_2
Right	↑	↓	Lower saturation; usually easier unloading.
Left	↓	↑	Higher saturation; usually harder unloading.

Use the exercise chain

Active tissue: CO_2 ↑ / H^+ ↑ / temperature ↑ → affinity ↓ → right shift → easier O_2 unloading.
P_{50} ↑ means more PO_2 is needed for half-saturation, so affinity is lower.

H+ cause comformational change of haemoglobin

Factors	Shift to	Explanation
↓pH/ ↑Pco_2	Right	High CO_2 indicates needs for O_2; CO_2 produces H^+, so: ↓pH → ↓affinity to unload O_2 (Bohr effect)
↑2,3-BPG#	Right	High 2,3-BPG indicates active metabolism, 2,3-BPG is acidic, so: ↓pH → ↓affinity to unload O_2 (Bohr effect) fetus hae is insensitive to 23BPG
↑temperature	Right	High temperature indicates active metabolism, so ↓affinity to unload O_2

#2,3-BPG = 2,3-bisphophoglycerate, products of glycolysis in erythrocytes

Original table with annotations, lecture p. 16. Read the mechanism clarification below.

Mechanism clarification: preserve the table, correct the interpretation

- Bohr effect: CO_2 ↑ / pH ↓ → reduced Hb affinity for O_2.
- 2,3-BPG binds and stabilises deoxyHb in the T state → right shift. It is not explained simply by "BPG is acidic", and is not itself the Bohr effect.
- [Extension] HbF binds 2,3-BPG less strongly → greater O_2 affinity and a left shift; "completely insensitive" is an overstatement.

Predicted practice P9 · Self-written; not an original exam question

"A higher P_{50} means that Hb has a higher affinity for oxygen."
Reasoning: False. A higher PO_2 is required for 50% saturation, indicating lower affinity.

13 / Curve questions: right shift and CO

Lecture PDF pp. 14–16 | Tutorial Q5 and Q10 | CO-specific mechanisms are extensions

Tutorial Q5 · Original tutorial question

A right shift of the oxygen dissociation curve reduces oxygen uptake by the lungs.

A condition-dependent statement: show both sides

- At fixed pulmonary PO_2, a right shift lowers Hb saturation and can reduce loading.
- In healthy lungs on the plateau, a moderate shift often has little loading penalty; enhanced tissue unloading is more prominent.
- A loading penalty becomes more important at low alveolar PO_2 or with a large shift.

Answer with the stated comparison: it is not correct to claim that right shifts never reduce loading. The question gives no PO_2 or shift magnitude, and no official key was supplied.

A concise English explanation

A right shift facilitates tissue unloading; its effect on pulmonary loading depends on alveolar PO_2 and the magnitude of the shift.

Tutorial Q10 · Original tutorial question

Carbon monoxide poisoning competes with O2 to bind to Hb. Therefore, it decreases Hb saturation and Hb affinity for O2.

Suggested answer: False; the affinity direction is wrong

CO competes for heme sites → fewer sites available for O_2 → effective carrying capacity and CaO_2 fall.
CO also increases the O_2 affinity of remaining available sites → left shift → impaired tissue unloading.
The danger is reduced carriage plus impaired release, not reduced affinity.

Measurement	Interpretation in CO poisoning
PaO_2	May remain normal: it reflects dissolved O_2, not CO occupancy of Hb.
Conventional SpO_2	May be falsely normal/high; a standard two-wavelength oximeter cannot reliably distinguish COHb.
“Saturation” terminology	Functional and fractional O_2 saturation differ when COHb is present. Do not equate a reassuring display with normal CaO_2.

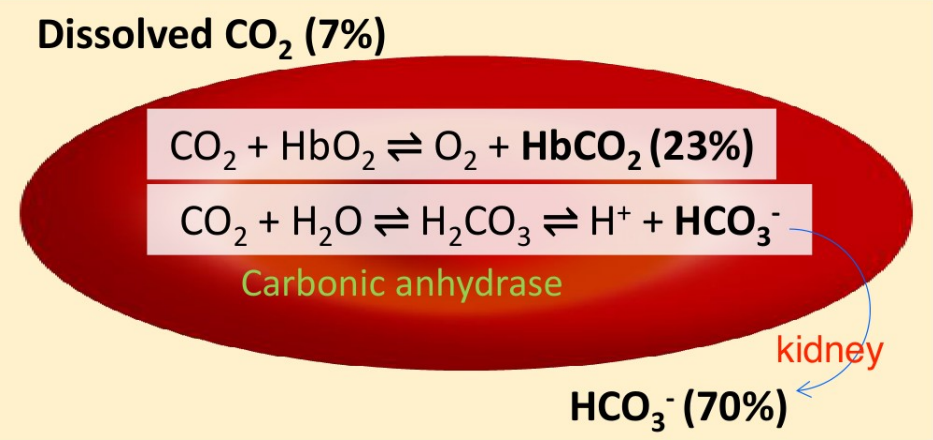
Predicted practice P10 · Self-written; not an original exam question

“A normal PaO_2 excludes carbon monoxide poisoning.”
Reasoning: False. CO binding to Hb need not lower dissolved O_2 pressure.

14 / CO_2 transport: three forms, one reaction

Lecture PDF p. 17

Transport form	Lecture estimate	What to remember
Dissolved CO_2	7%	Determines blood PCO_2.
Carbaminohemoglobin	23%	CO_2 binds globin amino groups, not heme Fe^{2+}.
Bicarbonate: HCO_3^-	70%: major form	Formed rapidly inside RBCs; much of the bicarbonate is then transported in plasma.



Original CO_2 transport diagram, lecture p. 17, including the "kidney" annotation.

Be able to reproduce the reversible reaction

$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$

Carbonic anhydrase (CA) catalyses reversible CO_2 hydration/dehydration.

It accelerates equilibration; it does not change the equilibrium constant.

Two similar names: do not mix them up

Name	Gas / binding site	Meaning
Carbamino-Hb: $HbCO_2$	CO_2 / globin amino groups	A normal component of CO_2 transport.
Carboxy-Hb: COHb	CO / heme sites	Impairs O_2 carriage and unloading; see p. 13.

Predicted practice P11 · Self-written; not an original exam question

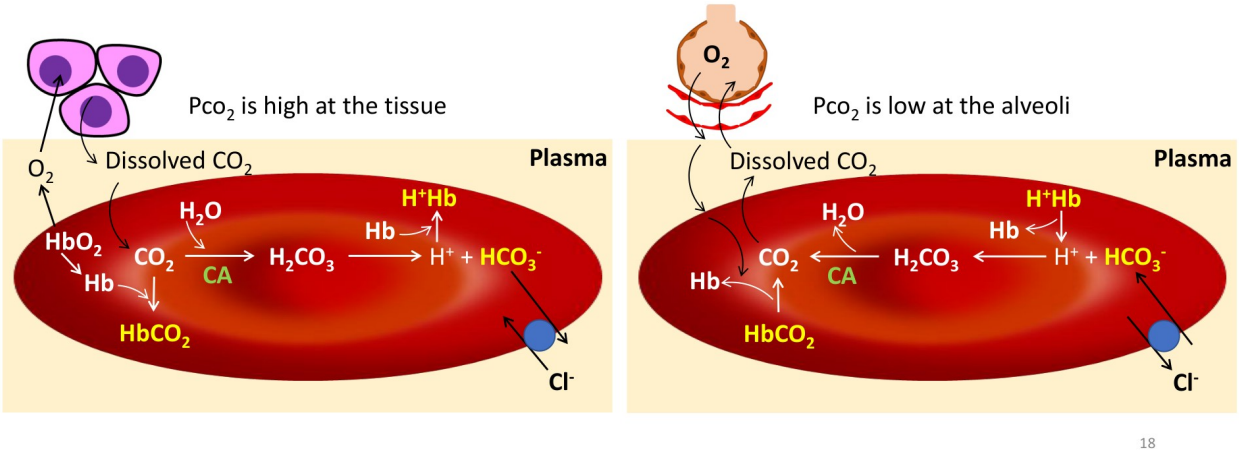
"Most CO_2 is transported bound to the heme iron of Hb."

Reasoning: False. Most is HCO_3^-; the carbamino fraction binds globin amino groups, not heme.

The 7%/23%/70% values follow this lecture; proportions vary slightly between sources and physiological conditions.

15 / Tissue loading and pulmonary unloading

Lecture PDF p. 18 | Chloride shift follows the directions shown in the original diagram



Original: systemic tissue end on the left, pulmonary end on the right; lecture p. 18.

Step	At tissue capillaries: load CO_2	At pulmonary capillaries: unload CO_2
1 Pressure gradient	High tissue PCO_2 → CO_2 enters blood and RBCs.	Lower alveolar PCO_2 → CO_2 diffuses into alveoli.
2 CA reaction	$CO_2 + H_2O \rightarrow H^+ + HCO_3^-$.	$H^+ + HCO_3^- \rightarrow CO_2 + H_2O$.
3 Hb buffering	After O_2 unloading, deoxyHb binds H^+ more readily.	O_2 binding promotes H^+ release from Hb.
4 Chloride shift	HCO_3^- leaves RBC; Cl^- enters to maintain electroneutrality.	HCO_3^- enters RBC; Cl^- leaves.
5 Carbamino transport	DeoxyHb favours CO_2 binding.	Oxygenation favours CO_2 dissociation from Hb.

Haldane effect: oxygenation is the input

DeoxyHb loads; oxyHb unloads

Tissues: Hb deoxygenation → greater CO_2 carriage and H^+ buffering → CO_2 uptake.
Lungs: Hb oxygenation → reduced binding of CO_2/H^+ → CO_2 release and exhalation.

Predicted practice P12 · Self-written; not an original exam question

"Oxygenation of Hb in the lungs increases its capacity to carry CO_2."
Reasoning: False. Oxygenation reduces CO_2/H^+ binding and facilitates CO_2 unloading.

Closed-book recall: four arrows

Tissues: CO_2 into RBC → HCO_3^- out, Cl^- in.
Lungs: HCO_3^- in, Cl^- out → CO_2 out of RBC → alveolus.

16 / Bohr versus Haldane: what changes what?

Lecture PDF pp. 16–18 | Tutorial Q12 partly overlaps; renal pharmacology below is an extension

Compare	Bohr effect	Haldane effect
Input	A change in CO_2 / H^+.	Hb oxygenation or deoxygenation.
Output	A change in Hb–O_2 affinity.	A change in Hb binding of CO_2 and H^+.
At tissues	CO_2 ↑ / pH ↓ → lower O_2 affinity → unloading.	DeoxyHb binds more CO_2 and buffers more H^+.
At lungs	CO_2 removal / pH recovery favours O_2 loading.	Oxygenation releases CO_2 and H^+ → CO_2 elimination.
Memory line	CO_2 / H^+ → O_2 transport.	Oxygenation → CO_2 transport.

The effects cooperate

Tissues need O_2: Bohr helps Hb release it; the resulting deoxyHb helps carry CO_2 via Haldane.
Lungs provide O_2: Hb loading promotes CO_2 unloading via Haldane.

Tutorial Q12 · Original tutorial question

Carbonic anhydrase inhibitor stimulates hyperventilation by promoting bicarbonate ion excretion.

Suggested answer: True, primarily through a renal mechanism

[Extension] A typical example is acetazolamide:
Renal tubular CA inhibition → less HCO_3^- reabsorption → more urinary HCO_3^- loss
→ metabolic acidosis → increased ventilatory drive.
At altitude, this can reduce the inhibition of ventilation caused by respiratory alkalosis.
Do not explain this as "CA inhibition makes RBCs produce more CO_2"; renal bicarbonate loss is the key.

The lecture directly covers CA in blood. Drug action, renal bicarbonate handling and ventilatory control require other lecture material; the extension supplies only the chain needed for Q12.

Predicted practice P13 · Self-written; not an original exam question

"An increase in tissue CO_2 facilitating O_2 unloading is the Haldane effect."
Reasoning: False. That is the Bohr effect. Deoxygenation improving CO_2 carriage is the Haldane effect.

17 / Integrated cases: test the causal chains

Self-written application cases | Cover the right column before answering

Case / question	Reasoning checkpoints
A | Hb is halved; PaO_2 and SaO_2 are normal. Which quantities change?	Capacity falls; CaO_2 is nearly halved. If cardiac output is unchanged, DO_2 falls. Saturation does not measure the amount of Hb. Links: Q3, Q6.
B | At altitude, FiO_2 stays constant and alveolar ventilation increases. What happens to PaO_2 and pH?	Relative to less ventilation at the same altitude: $PaCO_2$ ↓ → PAO_2 can rise. Acutely, pH ↑: respiratory alkalosis. PaO_2 need not return to its sea-level value. Links: Q4, Q7.
C | Compare fibrosis and emphysema. Which Fick variable changes?	Fibrosis: T ↑. Emphysema: A ↓. Both reduce diffusion capacity; cite the relationship $\dot{V} = AD\Delta P/T$.
D | CO_2 and temperature rise in working muscle. What happens to the curve and P_{50}?	Right shift; P_{50} ↑; affinity ↓; easier unloading. Deoxygenated Hb then favours CO_2 carriage. Distinguish Bohr from Haldane.
E | A diver descends to about 10 m without changing the gas mixture. What happens to N_2?	Absolute pressure approximately doubles → PN_2 ↑ → equilibrium dissolved N_2 ↑. Equilibration takes time. Link: Q13.
F | After CO exposure, PaO_2 and displayed SpO_2 appear normal. Does this exclude poor O_2 delivery?	No. CO occupies Hb sites and raises affinity at remaining sites → less O_2 carriage and impaired release. Conventional SpO_2 can mislead. Link: Q10.

Tell the tissue-end story in 60 seconds

A reference sequence

Muscle metabolism ↑ → CO_2 / H^+ / temperature ↑ → right shift → O_2 unloading
→ deoxyHb ↑ → stronger CO_2/H^+ binding (Haldane)
→ HCO_3^- out, Cl^- in → CO_2 transported towards the lungs.

Check whether your answer explains rather than labels

- Have you stated what stays constant: PO_2, cardiac output, temperature or gas fraction?
- Is each arrow attached to the correct quantity: pressure, saturation, content or delivery rate?
- Have you accidentally changed “can occur” into “must occur”?

18 / Final recall sheet and answer templates

Lecture PDF p. 19 and the preceding notes | Reconstruct this page from memory

When the question says...	Think...
Gas fraction unchanged	Check total pressure: Pgas = Fgas × Ptotal.
Dissolved gas changes	Henry: C = αP; specify temperature, medium and equilibrium.
Emphysema / fibrosis	A ↓ / T ↑; $\dot{V} = AD\Delta P/T$.
Normal saturation	Does not guarantee normal Hb, CaO_2 or DO_2.
P_{50} ↑ / right shift	Affinity ↓; easier unloading at the same tissue PO_2.
CO poisoning	Fewer available O_2 sites; higher affinity at remaining sites; left shift.
CO_2 transport	7% dissolved; 23% carbamino; 70% bicarbonate.
Bohr / Haldane	CO_2/H^+ affects O_2 affinity / Hb oxygenation affects CO_2 carriage.
Cyanosis and anemia	Consider absolute deoxyHb; no cyanosis does not exclude hypoxia.

Three useful exam-answer structures

1 | Counterexample: useful for "must" or "always"
False. Normal saturation does not guarantee normal oxygen delivery, because delivery also depends on haemoglobin concentration and cardiac output.

2 | Causal chain: useful for pressure and dissolved gas
False. At an unchanged nitrogen fraction, an increase in total pressure raises nitrogen partial pressure and therefore its equilibrium dissolved concentration.

3 | Conditional comparison: useful for curve questions
At the same PO_2, a right shift lowers Hb saturation. Its effect on pulmonary loading depends on PO_2 and the degree of the shift.

Final closed-book questions

- Why can an anemic person have normal PaO_2 and SaO_2 but inadequate oxygen delivery?
- Why does CO_2 cross the membrane readily despite its larger molecular weight?
- Why does Hb become better at carrying CO_2 after unloading O_2 at tissues?

Original-question locations: Q3 p. 9; Q4/Q7 p. 4; Q5/Q10 p. 13; Q6 p. 10; Q12 p. 16; Q13 p. 5. Return to the relevant other lectures for the remaining tutorial questions.